

NCFE Level 3 Data Course

Unit 1: Open Dataset Assessment

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Task 1: Local Government – Council Expenditure

1. Choose a local council

I chose Enfield.

2. Download its expenditure data

I went to the official .gov website for the London Borough of Enfield, and downloaded the publicly available council expenditure data. The data expenditure did not come in quarterly files, so I downloaded several months worth of data (May to August).

3. Determine the category with the highest spending

Based on the period where I selected data from, people services was the highest spending category, with a total of £101,995,515.32.

As the expenditure data did not come in one file, I had to concatenate and fit the data to my needs. Firstly, in order to extract the data I needed from the files, I loaded them into pandas with Python, dropping the columns I knew I didn't need. I also knew I needed to categorise all the types of data expenditure for each individual transaction, so I first created a dictionary of all the expenditure types by iterating through the file and appending any new strings to an array. Every time a transaction was iterated through, it would be appended to the corresponding dictionary entry.

After appending every transaction to the correct dictionary key, I appended all the key names to a new tuple and created a new dataframe, with column headings corresponding to the dictionary keys and creating new empty entries for the max number of rows required. I then finally inserted every object in each of the dictionary keys into the correct column based on the key and column heading. After appending every object, I wrote the new dataframe to an excel file. I did this for all the files from May to August.

After completing this for all files, I loaded them into a new Python file, before concatenating them into a new dataframe and converting the new dataframe to an excel file, allowing me to conduct the numerical analysis as instructed by the task.

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# NCFE Level 3 Data Unit 1: Task 1
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# =====

import pandas as pd
import numpy as np

#pd.set_option("display.max_rows", None)
#pd.set_option("display.width", None)

df = pd.read_excel("August.xlsx")

#df = pd.read_csv("May.xlsx", header = 0, sep = ",")

dfTrim = df.drop(["LONDON BOROUGH OF ENFIELD", "Payment Date", "Transaction
Number"], axis = 1)

print(dfTrim)
print(len(dfTrim))

dfTrimSet = set(dfTrim["Service Area Categorisation "])
print(dfTrimSet)

expenditureCategory = {}
dictLength = []

for i in dfTrimSet:

    expenditureCategory[str(i)] = []

print(expenditureCategory)

for i in range(len(dfTrim)) :

    x = dfTrim["Service Area Categorisation "][i]

    if x in expenditureCategory :

        #print(x)
        expenditureCategory[x].append(float(dfTrim["Net Amount"][i]))
        #print(i)

for i in expenditureCategory.keys():

    dictLength.append(len(expenditureCategory.get(i)))

print(dictLength)

newDf = pd.DataFrame()
newDf.insert(loc = 0, column = "debug", value = None)

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maxLength = max(dictLength)

for x in range(maxLength) :

    newDf.loc[x] = [None]

for ind, i in enumerate(expenditureCategory.keys()) :

    valueArray = []

    for index, x in enumerate(expenditureCategory.get(i)) :

        valueArray.append(x)

    newDf.insert(loc = ind + 1, column = i, value = pd.Series(valueArray))

#print(newDf)

newDf.to_excel("AugustK.xlsx", index = True)

# =====
# =====
# =====
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import pandas as pd

dfMay = pd.read_excel("MayK.xlsx")
dfJune = pd.read_excel("JuneK.xlsx")
dfJuly = pd.read_excel("JulyK.xlsx")
dfAugust = pd.read_excel("AugustK.xlsx")

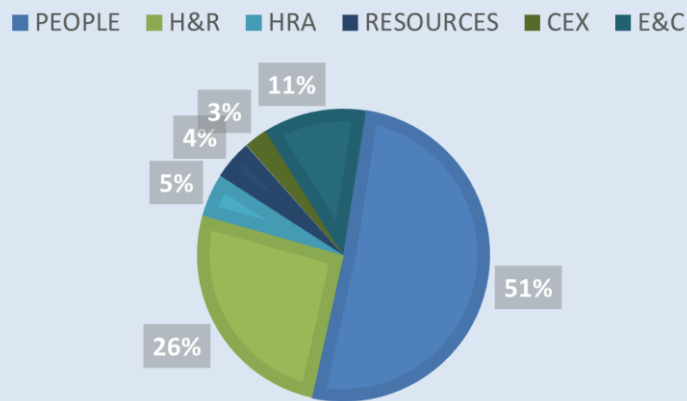
dfConcat = pd.concat([dfMay, dfJuly, dfJune, dfAugust])
dfConcat = pd.DataFrame(dfConcat)
#expenditureArray = [dfMay, dfJune, dfJuly, dfAugust]

dfConcat.to_excel("Test1P.xlsx", index = True)

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4. Visualise the top categories

TOTAL EXPENDITURE DISTRIBUTION, LB ENFIELD, MAY TO AUGUST



We can see from the pie chart that of all 6 categories, 'People' had just over half of all the council's total expenditure, with the subcategories contained within it being things such as Adult/Child Social Care, Education Services, and Health Support Services. This shows that a clear emphasis has been placed on services that directly impact people's wellbeing and education by the council. As these are services that are used by a large number of constituents on a daily basis, this makes a lot of sense. The next highest category is H&R, and by looking at the subcategories such as Construction Capital and Housing Repairs, we can see that it is clearly linked to the HRA category, which stands for Housing Revenue Account, and is linked to social housing maintained by the local council – specifically, H&R is Housing and Regeneration. It makes sense that these services would take up a large portion of council expenditure, as land prices and property management are very costly endeavours.

CEX is a category directly related to the costs associated with administration, the council's senior leadership, and associated corporate costs. The fact this is the lowest category with only 3% of total expenditure is a good sign that the council is being run efficiently. 11% of the total expenditure is on E&C with subcategories such as Waste Services and Vehicle Parts. E&C stands for Environment and Communities, and with more of a focus in recent times on creating a 'green' or environmentally friendly future, it is understandable that this category sees a reasonable chunk of spending.

By analysing this data and cross-checking the total expenditure in each category against the manifesto for each local governing body, we can either confirm or scrutinise whether the local council is upholding the budgets which they promised when they were elected. It is important to maintain a healthy interest and knowledge

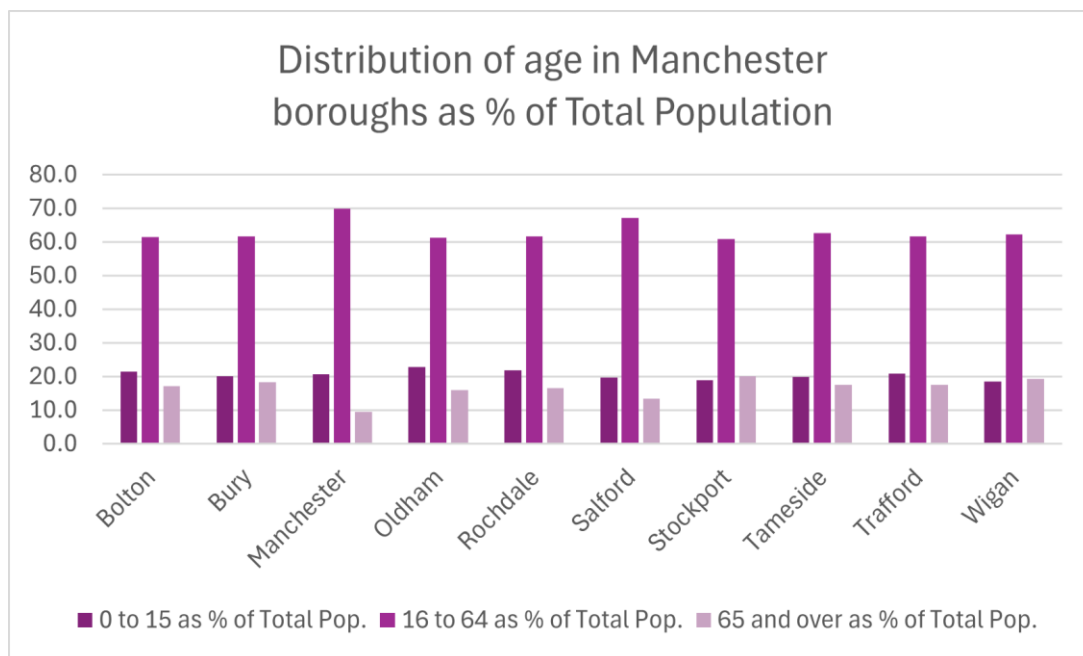
of what the council is doing with tax money, in case of any mismanagement due to negligence or even malicious intentions. We can also compare the spending with our own knowledge of what matters – and the likely costs of specific services – in order to better understand how well the council is utilising their available resources, and to identify areas that may either require an increase in the amount of resources available to them, or areas that could perhaps be cut down upon to allocate them somewhere more useful, making the council more efficient as a whole. There is also the ability to check the spending against past expenditure distribution, so we can see what focuses have changed for the government, and where spending is likely to head in the future.

Task 2: ONS – Census 2021

1. Download population data from Census 2021
2. Focus on the Greater Manchester Area

I downloaded the Population by sex and age group wide la gm csv file from the Trafford Data Lab that specifically pertains to the 2021 Census data for the Greater Manchester Area. The data itself came clean and complete, so there was no need to pre-process it.

3. Analyse age distribution across boroughs



I decided to convert the raw population data to the percentage of the total population in each borough in order to display the data as a clustered column chart. On the x-axis, we can see the different boroughs that make up the Greater Manchester area, and on the y-axis is the % of the total population of that area which is made up by each age group. I chose to represent the data like this, converting it to a percentage

of the total, rather than using the raw data as an input, as this is a better way to compare the actual distribution of age groups within each borough. If I left it as the raw data, it would be hard to see the relative number of people in, for example, the 16 to 64 age bracket in Manchester compared to Tameside, as the overall population of Manchester is much greater than that of Tameside, making it difficult to parse the data in visual format.

Based on this data, we can clearly see that Manchester and Salford have higher concentrations of people aged 16-64 years old – in other words, people of working age. Using our knowledge of Manchester being a metropolitan area, and Salford having a large working-class population with strong ties to the industrial field of employment, we can understand why this data makes sense. It also explains their relatively low percentage of people aged 65 and older, as for Manchester it would be difficult for those not working to maintain sufficient funds to live in the city, as well as the choice many make to move out of urban areas and into more rural ones after retirement. The large percentage of younger age groups in Oldham suggest that lots of families live there, and perhaps a higher level of affluence, as these could be families with parent who commute into the city for work whilst living in a more suburban, middle-class environment. Stockport and Wigan are the only two boroughs to have larger populations of 65 and over people than 15 and under, suggesting that many people move to this area when they retire, as their children or grandchildren are no longer present in the area.

It's important to keep track of the age distribution across different areas, so that local councils can better understand the needs of their constituents. For example, in an area like Manchester with a large working-age population, it's important to provide solid, reliable infrastructure in order to facilitate efficient business practices. For Oldham with a large number of young people, education should be a priority, as there will be lots of schoolchildren in the area. In Stockport and Wigan with large elderly populations, the council should ensure a strong healthcare service, to make it readily available for those that need it.